

SHUNTING

Preparation & Berthing

Revision Guide — EMR Training / GERT8000-SS2 Issue 7

⚠ All rule book content sourced exclusively from GERT8000-SS2 Issue 7 ⚠

1. Key Definitions

Term	Definition
Shunting movement	Any movement of a train or vehicle OTHER than a train passing normally along a running line.
Shunter	The person in control of a specific shunting movement.
Unaccompanied driver	A driver carrying out a shunting movement without an accompanying shunter.
Propelling	Pushing vehicles by a traction unit. Does NOT include push-pull trains.
Loose shunting	Shunting of vehicles that do not remain attached to the traction unit during the movement.
Points worked from a signal box	Includes points worked from a ground frame.
Tandem working	Two or more locos driven by two or more separate drivers.
Multiple working	Two or more locos driven by ONE driver through control cables connecting all locos.
Foul	A vehicle is "foul" when it is partially obstructing an adjacent line or fouling point.

2. Depots, Sidings and Sources of Movement Authority

2.1 What are Depots and Sidings?

Depots, sidings and stabling points are where trains are stabled prior to coming into or after service. Some have maintenance facilities. Unlike running lines, movements are more typically controlled by verbal instructions. Some areas use position-light signals; others rely entirely on verbal commands, hand signals, or stop boards.

2.2 Sources of Movement Authority

Authority to make a shunting movement can come from several sources:

- The controlling signaller clearing a signal onto a reception road or siding.
- A shunter — at a stop board you must stop and wait for instruction; at a position-light signal you obey the aspect.
- A panel operator — may give details of the move by radio before clearing a signal.
- A driver diagram — for shunt movements on the main line that do not require a shunter (unless entering a yard or siding).

The shunter is IN CHARGE of specific shunting movements. You must work to the instructions given — even if a signal has cleared for the movement.

2.3 Identifying Your Stock

Before navigating to your stock you must locate and identify its berthing position. This information will be on your duty diagram, schedule card or movement authorisation, or given verbally by an authorised person (e.g. depot supervisor or shunter).

- Within depots/sidings — stock location may be given as a "road" number or a description.

- At stations — communicated by platform number, headcode, destination, or arrival/departure time.

Always confirm the road number AND unit number before boarding. Boarding the wrong train is a serious risk.

3. Train Preparation

3.1 Types of Preparation

Type	Description
Full Preparation	All required exterior, cab and interior checks carried out.
Partial / Mobilisation	Basic check of traction, braking, communication and door systems only.
Above Solebar / Station Preparation	Used where trains are berthed in platforms and exterior equipment cannot be checked. Interior and cab checks only.

3.2 What Train Preparation Covers

Train preparation is the process of systematically checking the train/rolling stock is in a safe and working condition. It covers:

- Mechanical components.
- Lights, wipers and warning horns.
- Traction and braking systems.
- Fluid levels.
- Safety systems (including AWS/TPWS).
- Communication systems.
- Emergency equipment.
- Cleanliness and condition of the train (interior and exterior).

Preparation tasks are found in Traction Manuals — they act like an airline pilot's checklist. Perform them systematically and thoroughly.

3.3 Reporting Missing or Defective Items

One of the main purposes of train preparation is to identify missing or defective items before the train enters service. Report any such items to the relevant person — typically the maintenance department, depot supervisor or operations controller.

Risks of non-compliance or taking shortcuts include: train failure, delays, operational incidents, death or injury, financial costs, disciplinary action and legal proceedings.

3.4 Electrical Safety

Owing to the various electrical voltages on any form of motive power, you must consider any electrical system to be LIVE at all times.

3.5 Safe Access and Egress

When boarding a traction unit:

- Put your bag in first.
- Use both handrails and climb up using each step, maintaining 3 points of contact.
- Ensure you are safely inside before picking up your equipment.

When leaving a traction unit:

- Make sure it is safe to leave — check for train movements on adjacent lines.
- Beware of CRE (current-related equipment) and uneven surfaces (e.g. ballast).
- Place your equipment on the floor before climbing out.
- Locate both handrails and descend backwards, finding each step and maintaining 3 points of contact.
- Do NOT jump from the vehicle or miss steps — this can cause serious injury.

4. Prohibitions and Restrictions (SS2 \$2)

4.1 Starting a Shunting Movement

Unless authorised, you must start a shunting movement using a traction unit.

You may only move vehicles using a chain, rope or road vehicle where this is SPECIALLY AUTHORISED in local instructions. You must NEVER move vehicles using a prop or pole against a locomotive or any rail or road vehicle.

4.2 Loose Shunting

Loose shunting (vehicles becoming detached from the traction unit during the movement) may only be carried out where SPECIALLY AUTHORISED in local instructions.

You must NOT loose shunt coaching stock vehicles. You must NOT loose shunt other vehicles against coaching stock vehicles.

5. Precautions Before Shunting (SS2 \$4)

5.1 Reaching a Clear Understanding (\$4.1)

Before starting any shunting, the driver and shunter must reach a clear understanding about:

- Exactly what needs to be done.
- How the shunting movements will be controlled.

5.2 Safety Checks Before Any Movement (\$4.2)

The shunter (or unaccompanied driver) must make sure that:

- The vehicles can be moved safely.
- No NOT TO BE MOVED boards are placed on the vehicles.
- Other vehicles are not foul of the movement to be made.

- Any road vehicle or equipment is clear.
- Anyone who could be put in danger is warned to move to a safe position.
- Anyone working on the outside of vehicles on an adjacent line is warned to keep clear.
- Any derailer or scotch block has been removed.
- All points the movement will go over are correctly set for the movement (unless trailing hand points are specifically authorised to be pushed through).

6. Safeguards While Shunting (SS2 \$5)

6.1 General Rules (\$5.1)

The driver must work ONLY to the shunter's instructions.

A movement must only be made, even when a signal has been cleared, if the shunter has:

- Authorised the movement, OR
- Operated a shunting or other indicator which authorises the movement.

SPEED LIMIT: You must not exceed 5 mph (10 km/h) in a siding unless specifically authorised.

Shunting movements must NOT pass a signal at danger, a block marker or shunt marker without authority.

6.2 Controlling Movements by Hand Signal (\$5.2a)

The shunter must:

- Use the prescribed handsignals only.
- Make sure the driver can see the handsignals at all times.
- Make sure no other driver acts on the handsignals.

The driver must:

- NOT start or continue a movement unless they clearly understand the handsignal.
- NOT start or continue unless they are sure the handsignal applies to them.
- Stop IMMEDIATELY if they lose sight of the shunter or the shunter's handsignals.
- Restart only when the shunter gives the correct handsignal.

Handsignal	Meaning
Arm(s) moved in circular motion / white light waved in circular motion	Move away (away from shunter)
Arm(s) moved towards body / white light moved slowly towards body	Move towards (towards shunter)
Arm raised, moved up and down / light raised and moved up and down	Ease up (slow down)
Arm(s) raised above head / red light raised	Stop
One arm held steady (in vehicle)	Stop when in a vehicle

6.3 Controlling Movements by Radio (\$5.2b)

When using radio, the shunter must:

- Clearly identify the correct train and driver (use call signs, e.g. "Shunter 1 to Driver 1").
- Speak CONTINUOUSLY or transmit a CONTINUOUS BLEEP SIGNAL throughout each movement.
- Instruct the driver to stop immediately if transmission is failing.

The driver must:

- Stop IMMEDIATELY if there is a break in transmission.
- Restart only when the shunter confirms it is safe to continue.

If there is a break in radio transmission — STOP immediately. Restart ONLY when the shunter has confirmed you may continue.

6.4 Bell/Buzzer Codes

Bells/Buzzes	Meaning
1	Stop
2	Move away (in direction of travel)
3	Move towards (reverse)
4	Ease up / slow down
6	All clear / right away

6.5 Speed Limits During Shunting

Location	Speed Limit
Maintenance depot, servicing or stabling siding	5 mph (unless authorised higher)
Washer, wheel lathe, sheds, fuel lines (F&I)	3 mph
Sidings generally (SS2 §5.1)	5 mph (10 km/h)

6.6 Propelling (§5.3)

When a movement is not driven from the leading cab (i.e. propelling), the shunter must:

- Ride in the leading cab (if at the leading end) and be able to apply the automatic brake, OR
- Ride in the vehicle at the leading end (if suitable) and be able to control the movement and apply the automatic brake, OR
- Control the movement from a safe place on the ground, AHEAD of the movement, in contact with the driver or where the driver can see the shunter.

During the propelling movement the shunter must keep a good lookout, obey all signals, warn anyone on or near the line, and stop if anyone appears to be in danger.

Propelling is only allowed if necessary and must be over the shortest possible distance.

6.7 Driving from the Leading Cab (SS2 §6)

You must always drive from the leading cab when a shunting movement is:

- Within a depot or stabling siding.
- Entering a shed or building.
- Proceeding onto vehicles.

- Approaching buffer stops.

You may drive from another cab ONLY if a shunter is controlling the movement by radio AND it is not necessary to observe signals or handsignals.

6.8 Entering a Shed or Building (\$5.7)

Before allowing a movement to enter a shed or building you must:

- Stop the movement at the entrance.
- Only proceed when it has been checked it is safe to do so.
- Sound a short blast of the horn (soft setting or depot whistle) as a warning before restarting — unless otherwise authorised.

6.9 After Each Movement (\$5.4)

After each movement the shunter/unaccompanied driver must:

- Make sure vehicles are secured by handbrakes where necessary.
- Scotch vehicles which have no handbrake or on which the handbrake is not working (unless attached to other vehicles with working handbrakes that can hold them).
- Secure stationary vehicles before making a draw-away movement.

You must NOT rely on the automatic brake to secure any vehicle.

6.10 Limit of Shunt / Shunting Beyond a Home Signal (\$5.5 & \$5.6)

The shunter/unaccompanied driver must ensure no part of the movement passes a limit of shunt signal or indicator unless the signaller has given permission.

A wrong-direction shunting movement must not go beyond a home signal without the signaller's permission. If there is a falling gradient towards the next signal box, the movement must not be made unless:

- The automatic brake is working throughout the train, OR
- The locomotive is at the end nearer to the next signal box.

An overrun of a Limit of Shunt board is considered a SPAD (Signal Passed at Danger).

7. Points

Type of Points	Description
Facing points	The train approaches the toe of the points — the direction of travel is determined by which way the points are set.
Trailing points	The train approaches from the heel — the points are pushed aside by the flanges of the wheels.
Hand points	Manually operated points that must be set correctly for the movement.
Motorised points	Points operated electrically by signal box or ground frame.
Trap points	Points designed to derail a runaway vehicle to prevent it fouling a main line.

Catch points	Similar to trap points — protect against runaways on a gradient.
Spring points	Points held in a normal position by a spring; can be pushed through in one direction but spring back automatically.

You must NOT trail through unset hand points (unless approved at that location). You must clear any hand points with the FULL TRAIN before turning back, unless you are sure they are set correctly.

7.1 Movements Over Points Worked from a Signal Box (SS2 §8)

Before authorising a movement over signal-box-worked points, the shunter/unaccompanied driver must get the signaller's permission (verbally or by handsignal) and check the points are fitting correctly where possible.

Signaller's permission signal:

- Daylight — arm raised above the head.
- Darkness — white light twisted quickly.

Permission is NOT needed if the signaller has cleared a signal for the movement, or if the movement passes a shunting/position-light signal with a yellow stop indication and the points are set for a route to which the signal does not apply.

When the movement is clear of points, the shunter/driver must indicate this to the signaller:

- Shunter (daylight) — arm raised above the head.
- Shunter (darkness) — white light twisted quickly.
- Unaccompanied driver (any time) — one short blast on the horn.

8. Attaching and Detaching Vehicles (SS2 §7)

8.1 Approaching to Couple (§7.2)

When attaching a traction unit to a train or vehicles you must:

- Always stop the traction unit 2 metres (6 feet 6 inches) from the vehicle.
- Stop again at any distance set out in the instructions for the class of traction unit.
- If controlled by a shunter, move forward only when authorised by the shunter.

Ensure both coupling heads are in alignment. If a "NON MULTI" sticker is displayed, do not attempt to couple. Some traction requires a brake continuity test once coupling has taken place.

8.2 Detaching — Before Uncoupling (§7.3)

Before detaching a traction unit:

- Secure the train. If on a gradient, secure it at the LOWER end.
- Do NOT rely on the automatic brake to secure the train.

Before detaching a vehicle from a train, secure the vehicle. Before a dead traction unit is detached, make sure it is properly secured.

8.3 Uncoupling on a Running Line (§7.4)

You must NOT uncouple a traction unit from another traction unit on a running line except:

- At a signal box.
- At a signal.
- On a platform line.

Before uncoupling at a location where this does not happen routinely, you must tell the signaller what movements need to be made.

8.4 Passenger and Postal Trains (§7.1)

The automatic brake must be in use on movements which involve attaching to or detaching from a passenger or postal train.

9. Completing Shunting & Securing Vehicles (SS2 §9)

9.1 Leaving Vehicles in a Safe Position (§9.1)

When shunting is completed you must make sure vehicles are:

- Not left on a running line (except as permitted in §9.3).
- Not fouling any other line.
- Clear of any points which need to be moved.
- Left within the protection of any trap points, derailleurs or scotch blocks.
- Leaving enough room at fouling points for anyone to pass safely between the vehicles and any movement on the adjoining line or siding.

9.2 Securing Vehicles (§9.2)

Vehicles and traction units must be properly secured to prevent movement. Normally this means applying the parking/handbrake. If the handbrake is isolated or inoperable, liaise with the shunter or engineering staff to confirm immobilisation by: handbrake, attaching to another vehicle, or use of SCOTCHES.

You must NOT rely on the automatic brake to secure any vehicle.

9.3 Leaving Vehicles on a Running Line (§9.3)

When leaving vehicles on a running line you must:

- First tell the signaller (unless the method of working is routine at that location).
- Place a RED light on the REAR end of the vehicles, or on BOTH ends if on a single or bi-directional line.

9.4 Leaving Vehicles on a Dead-End Line (§9.4)

When leaving vehicles on a dead-end line which has a red or white light on the buffer stops, you must make sure a light of the SAME COLOUR is placed on the end of the vehicles which faces approaching movements.

You do not have to place a light if your TOC instructions say it is not necessary at that location because the vehicle will be sufficiently visible to an approaching movement.

9.5 Protecting Running Lines (\$9.5)

To protect running lines after shunting, make sure that:

- Ground-frame operated points and derailleurs are left in the normal position.
- Scotch blocks, where provided, are placed across the rails.

9.6 Decommissioning and Berthing

When immobilising rolling stock the basic principles are:

- Rolling stock secured to prevent movement.
- Diesel engines shut down to prevent fuel wastage and noise pollution.
- Auxiliary equipment (e.g. lighting) switched off to prevent battery discharge.
- Exterior lighting configured as required in local working instructions.
- Train cab decommissioned and driver's/master key removed.
- Cab doors secured and windows closed to prevent unauthorised access.

After berthing, report the train's position to the person in control of movements — the shunter or movements controller on a depot, or the controlling signaller for platform lines or sidings.

10. Foul Vehicles and Risk Management

You must NOT leave a train foul without permission. If you do leave it foul, you must tell the shunter.

Before any movement, you must be satisfied that:

- Points are lying in the correct position for the movement.
- No other vehicle is standing foul for the movement.
- No conflicting train movements are taking place.
- Any signals for the movement are cleared.
- Anyone on or near the line is aware of the pending movement.
- Any cables or connections attached are removed.
- For movements into a maintenance shed — the required announcement or warning alarm has been activated, shed doors are correct, and the designated person has given authority.

If detained on a running line for an unusually long time, remind the signaller of your presence in the quickest way possible — if necessary by going to the signal box personally or sending the shunter.

11. Questions and Answers

Q1. What is a shunting movement?

A1. Any movement of a train or vehicle other than a train passing normally along a running line.

Q2. What is the difference between tandem working and multiple working?

A2. Tandem: two or more locos driven by two or more drivers. Multiple: two or more locos driven by ONE driver through control cables.

Q3. What is propelling?

A3. Pushing vehicles by a traction unit. This does NOT include push-pull trains.

Q4. What is loose shunting and when is it permitted?

A4. Shunting where vehicles become detached from the traction unit during the movement. Only permitted where specially authorised in local instructions. Must NEVER be used with coaching stock.

Q5. Who is in charge of a specific shunting movement?

A5. The shunter. The driver must work to the shunter's instructions — even when a signal has cleared for the movement.

Q6. What must the driver and shunter agree before any shunting takes place?

A6. Exactly what needs to be done, and how the shunting movements will be controlled.

Q7. List the safety checks that must be carried out before any shunting movement.

A7. (1) Vehicles can be moved safely. (2) No 'NOT TO BE MOVED' boards on vehicles. (3) No vehicles foul of the movement. (4) Road vehicles/equipment clear. (5) People in danger warned to move to safety. (6) Anyone working on adjacent vehicles warned to keep clear. (7) Any derailer or scotch block removed. (8) All points correctly set for the movement.

Q8. What is the speed limit in a siding?

A8. Must not exceed 5 mph (10 km/h) unless specifically authorised.

Q9. What speed limits apply in washers, wheel lathes, sheds and fuel lines?

A9. 3 mph.

Q10. A signal has cleared for a shunting movement. Must the driver make the move?

A10. No. The driver must only make a movement if the shunter has authorised it OR operated a shunting/other indicator that authorises it — even when a signal is cleared.

Q11. The driver loses sight of the shunter's handsignals. What must the driver do?

A11. Stop the movement immediately. Restart only when the shunter has given the correct handsignal.

Q12. Radio communication breaks during a shunting movement. What must the driver do?

A12. Stop immediately. Restart only when the shunter tells the driver it is safe to continue.

Q13. When controlling a movement by radio, what must the shunter transmit throughout each movement?

A13. Speak continuously OR transmit a continuous bleep signal throughout each movement.

Q14. What does 1 bell/buzz mean during shunting?

A14. Stop.

Q15. What does 2 bells/buzzes mean?

A15. Move away (in direction of travel).

Q16. What does 3 bells/buzzes mean?

A16. Move towards (reverse).

Q17. What does 4 bells/buzzes mean?

A17. Ease up / slow down.

Q18. What does 6 bells/buzzes mean?

A18. All clear / right away.

Q19. What must happen before a movement enters a shed or building?

A19. Stop at the entrance. Check it is safe. Sound a short blast of the horn (soft setting or depot whistle) before restarting — unless otherwise authorised.

Q20. When must you always drive from the leading cab during shunting?

A20. Within a depot or stabling siding, entering a shed or building, proceeding onto vehicles, or approaching buffer stops.

Q21. Under what condition can a driver drive from a non-leading cab?

A21. Only if a shunter is controlling the movement by radio AND it is not necessary to observe signals or handsignals.

Q22. When can propelling take place and over what distance?

A22. Only if necessary, and must be over the shortest possible distance. The shunter must be in the leading cab, at the leading end, or ahead of the movement on the ground in contact with the driver.

Q23. What is the rule about a limit of shunt board?

A23. No part of the movement may pass a limit of shunt signal or indicator without the signaller's permission. An overrun is considered a SPAD.

Q24. What is the rule about leaving a train foul?

A24. You must not leave a train foul without permission. If you do leave it foul, you must tell the shunter.

Q25. Can the automatic brake be used to secure a vehicle at a stand?

A25. No. You must NEVER rely on the automatic brake to secure a vehicle.

Q26. What must you do when leaving vehicles on a running line?

A26. Tell the signaller (unless routine at that location). Place a red light on the rear end, or on both ends if on a single or bi-directional line.

Q27. What colour light must be placed on vehicles left on a dead-end line?

A27. The same colour as the light on the buffer stops (red or white) — placed on the end facing approaching movements.

Q28. When detaching a traction unit on a gradient, where must you secure the train?

A28. At the lower end.

Q29. Where can a traction unit be uncoupled from another traction unit on a running line?

A29. Only at a signal box, at a signal, or on a platform line.

Q30. Must the automatic brake be in use when attaching or detaching from a passenger train?

A30. Yes. The automatic brake must be in use on all movements involving attaching to or detaching from a passenger or postal train.

Q31. When approaching to couple, where must the traction unit first stop?

A31. 2 metres (6 feet 6 inches) from the vehicle, and then stop again at any additional distance specified for that class of traction.

Q32. What are the three types of train preparation?

A32. Full preparation (all checks). Partial/mobilisation (basic traction, braking, comms and doors). Above solebar/station preparation (cab and interior only — used when exterior cannot be accessed).

Q33. What must you confirm before boarding a train during preparation?

A33. The road number AND the unit number — to ensure you are not boarding the wrong train.

Q34. What are the potential consequences of taking shortcuts during train preparation?

A34. Train failure, delays, cancellations, operational incidents, death or injury, financial costs, disciplinary action, and legal proceedings or prosecution.